

Resources: calculator

Max Qiu is curious about how ionic bonds form. From information he found online, he knew that at the microscopic level, gravitational forces are generally not considered. He finds the problem interesting and decides to share it with Sean. Sean posits that a bond can only form stably when the total potential field of two particles sits at a local minimum. After more research, they find that two forces act between the two particles:

1. The Coulomb force

$$F_E(r) = k \frac{q_1 q_2}{r^2},$$

where r is the separation of the two particles and k is the Coulomb constant.

2. A short-range “resistance” force

$$F_R(r) = \frac{nA}{r^{n+1}},$$

which is repulsive and dominates at small r .

Here $n > 1$ and $A > 0$ are constants.

Assumptions

1. The two atoms are modeled as point particles of opposite charge, $\text{sgn}(q_1) = -\text{sgn}(q_2)$.
2. Only the Coulomb force and the short-range resistance force act; gravity and all other interactions are neglected.
3. The interaction is central — each force depends only on the separation r of the two particles.

Questions.

- Q1. Gravity versus Electricity.** Consider two electrons, each of charge $-e$ and mass m_e . Compute the ratio $|F_G|/|F_E|$ of the gravitational and electric forces between them, evaluate it numerically, and take its logarithm to explain why Max decided that gravitational forces are negligible.
- Q2. Potential Energy.** Given q_1 , q_2 , m_1 , m_2 , and the two forces $F_E(r) = k q_1 q_2 / r^2$ and $F_R(r) = nA / r^{n+1}$ (with k , n , A known constants), compute the total potential energy $U(r)$. Use $U(r \rightarrow \infty) = 0$ as the reference point.
- Q3. Stable Bond.** When a bond forms stably, the equilibrium separation r_0 and binding energy $U_0 = U(r_0)$ satisfy $\frac{dU}{dr}|_{r_0} = 0$ and $\frac{d^2U}{dr^2}|_{r_0} > 0$. Find r_0 and U_0 in terms of the given constants, and prove that the minimum is unique.